MIDTERM COMPUTATION REVIEWER

CS 402

Modelling and Simulation

Linear Models, Non-Linear Models, and Simulation Math

A clear, worked reviewer based on the computations shown in the Midterm Week 01 and Week 02-03 slides.

What this reviewer covers

1. Constant-speed distance and the meaning of a linear model
2. Using $y = mx + b$, finding slope, and solving for an unknown
3. First differences, changing rates, and $D(x) = x^2$
4. Unity computations: unit conversion, clamping, and smooth following
5. Practice problems with a complete answer key

Study tip: Cover each solution, solve the problem yourself, then compare every substitution and unit.

START HERE

Quick Formula Sheet

distance = speed x time

Use when the object starts at position 0.

y = mx + b

Output = rate x input + initial value.

m = (y2 - y1) / (x2 - x1)

Slope is the change in output divided by the change in input.

D(x) = x^2 = x times x

The output is the input multiplied by itself.

speed in m/s = speed in km/h / 3.6

Unity physics uses meters and seconds.

How to Decide Which Computation to Use

What the problem gives	What to do
Speed and time; starts at 0	Use distance = speed x time.
Rate plus a starting value	Use y = mx + b.
Two points or a table	Compute slope with change in y / change in x.
A table with equal x-steps	Find first differences in y.
Constant first differences	Classify as linear.
Changing first differences	Classify as non-linear.
A square rule	Substitute into D(x) = x^2.

Unit check

The rate's denominator must match the input. Example: meters/second must be multiplied by seconds.

WEEK 01

1. Linear Models: Constant Rate of Change

A relationship is linear when equal changes in the input always produce equal changes in the output. Its graph is a straight line.

$$y = mx + b$$

Symbol	Meaning	In a car-position problem
y	Dependent variable or output	Position of the car
x	Independent variable or input	Elapsed time
m	Slope or constant rate	Speed
b	y-intercept or initial value	Starting position at x = 0

Fast Interpretation

- $m > 0$: the output increases as x increases.
- $m < 0$: the output decreases as x increases.
- b is the value of y when $x = 0$.
- The units of m are output units per input unit, such as meters per second.

Common mistake

Do not forget b. The formula distance = speed x time assumes the starting position is 0. If the object already started somewhere else, add that initial value.

2. Worked Example: 60 km/h Car

A car travels at a constant speed of 60 kilometers per hour and starts at 0 kilometers. Find the distance after 1, 2, 3, and 5 hours.

distance = speed x time

Step-by-Step

1. Identify the rate: speed = 60 km/h.
2. Use time in hours because the rate is per hour.
3. Multiply 60 by each time value.

Time	Substitution	Distance
1 h	60 km/h x 1 h	60 km
2 h	60 km/h x 2 h	120 km
3 h	60 km/h x 3 h	180 km
5 h	60 km/h x 5 h	300 km

Why it is linear

Every additional hour adds exactly 60 km. The first difference is always +60 km.

Fresh Example

A conveyor moves products at 12 items/minute. After 7 minutes: output = 12 x 7 = 84 items. The same linear idea works outside transportation.

3. Worked Example: Position With an Initial Value

At 8:00 AM, a car is already 100 meters from a checkpoint. It then moves at 20 meters per second. Find its position after 5, 10, 15, and 20 seconds.

y = 20x + 100

m = 20 m/s and b = 100 m

One Complete Substitution

1. Choose x = 5 seconds.
2. Substitute: y = 20(5) + 100.
3. Multiply: y = 100 + 100.
4. Add the initial position: y = 200 meters.

Time x (s)	20x	+ 100	Position y (m)
0	0	+ 100	100
5	100	+ 100	200
10	200	+ 100	300
15	300	+ 100	400
20	400	+ 100	500

Sanity check

At x = 0, y must equal 100. Every additional second must add 20 meters.

4. Finding Slope, Intercept, and Missing Values

A. Find slope from two points

Using (0, 100) and (20, 500):

m = (500 - 100) / (20 - 0) = 400 / 20 = 20 m/s

B. Find the intercept

The intercept is the output when x = 0. From the table, y = 100 at x = 0, so b = 100 meters.

C. Solve for time

When will the car reach 700 meters?

1. Start with 700 = 20x + 100.
2. Subtract 100: 600 = 20x.
3. Divide by 20: x = 30 seconds.

D. Build a model from words

A drone begins 40 meters from home and travels away at 12 m/s.

y = 12x + 40

After 7 seconds: y = 12(7) + 40 = 84 + 40 = 124 meters.

Exam habit

Write the model before substituting numbers. It makes the role of every number visible and reduces sign errors.

WEEK 02 AND 03

5. First Differences: Linear or Non-Linear?

When x increases by equal amounts, subtract consecutive y-values. These results are the first differences.

first difference = next output - current output

Traffic Delay Table From the Slides

Cars	Delay	First difference
100	5 min	-
200	7 min	7 - 5 = +2
300	11 min	11 - 7 = +4
400	17 min	17 - 11 = +6
500	25 min	25 - 17 = +8

Conclusion

The x-step is always +100 cars, but the delay changes by +2, +4, +6, and +8 minutes. Because the rate is changing, the relationship is non-linear.

Compare With a Linear Table

Cars x	Delay y	First difference
1	2	-
2	4	+2
3	6	+2
4	8	+2
5	10	+2

6. Quadratic Example: $D(x) = x^2$

The rule x^2 means multiply x by itself. This is non-linear because equal x-steps do not create equal first differences.

x	Substitution	$D(x) = x^2$	First difference
1	1 x 1	1	-
2	2 x 2	4	+3
3	3 x 3	9	+5
4	4 x 4	16	+7
5	5 x 5	25	+9

Why the Changes Are 3, 5, 7, 9

4 - 1 = 3; 9 - 4 = 5; 16 - 9 = 7; 25 - 16 = 9. The first differences change, so the function is non-linear.

Second Differences

Subtract the first differences: 5 - 3 = 2, 7 - 5 = 2, and 9 - 7 = 2. A constant second difference is a strong clue that the model is quadratic.

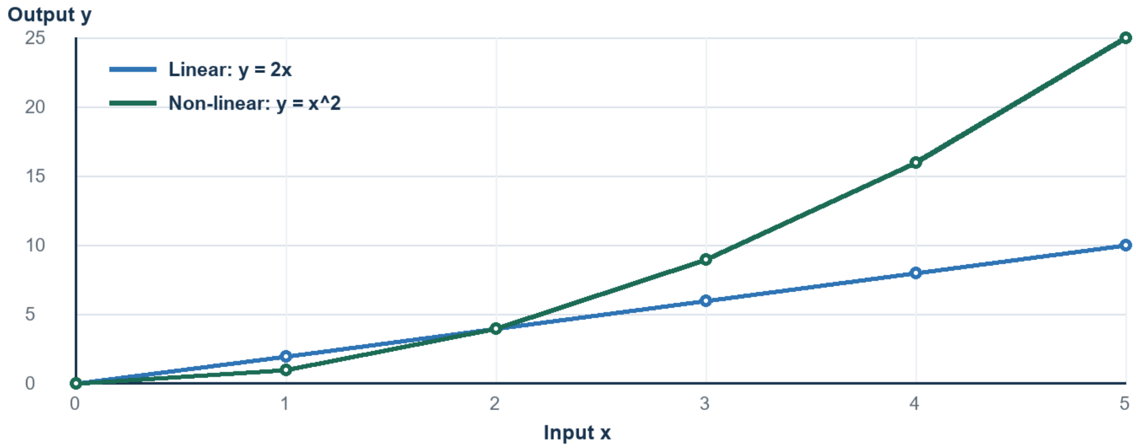
Fresh Example

Find D(6): $D(6) = 6^2 = 6 \times 6 = 36$. The increase from D(5) to D(6) is 36 - 25 = 11, continuing the odd-number pattern.

Do not confuse the two traffic examples

The 100-to-500 car table demonstrates changing differences. The separate $D(x) = x^2$ table demonstrates a square rule. They are both non-linear examples, but they do not use the same x-scale.

7. Linear vs. Non-Linear at a Glance



Equal x-steps create constant changes for $y = 2x$ and increasing changes for $y = x^2$.

Test	Linear	Non-linear
Rate of change	Constant	Changes
First differences	Constant	Not constant
Typical graph	Straight line	Curve or another non-straight shape
Example	$y = 2x + 3$	$y = x^2$

Three-Step Classification Check

1. Confirm that the x-values increase by equal steps.
2. Subtract consecutive y-values.
3. Constant differences mean linear; changing differences mean non-linear.

8. Simulation Computations in the Unity Slides

A. Convert km/h to m/s

The CarMovement script uses speedMS = speedKPH / 3.6 because Unity's distance unit is treated as a meter and physics time is in seconds.

60 km/h / 3.6 = 16.67 m/s

At this constant speed, the car travels about 16.67 meters in 1 second and about 83.33 meters in 5 seconds.

B. Velocity vector

new Vector3(speedMS, rb.velocity.y, rb.velocity.z) sets the x-speed to the converted value while preserving the current y- and z-components. The scripted motion is mainly along the x-axis.

C. Clamp the road count

RoadNumber = Mathf.Clamp(RoadNumber, 0, Roads.Length) keeps the requested count within the array capacity. If Roads.Length = 99 and RoadNumber = 120, the clamped value is 99.

D. Smooth following with interpolation

The camera uses Lerp for position and Slerp for rotation. The blend fraction is approximately smoothing x Time.deltaTime.

blend = 5 x (1/60) = 0.0833

At 60 frames/second, the camera closes about 8.33% of the remaining gap per frame.

If the positional gap is 12 meters, the first correction is about 12 x 0.0833 = 1 meter. The exact motion eases because the remaining gap becomes smaller each frame.

Important distinction

The linear-model formula predicts position from a constant rate. Lerp produces smooth easing toward a target, so its step size changes as the gap changes.

9. Common Mistakes and Fast Fixes

Mistake	Why it is wrong	Fast fix
Using $y = mx$ and ignoring b	The object may not start at 0.	Check the value at $x = 0$.
Mixing seconds with km/h	The time unit does not match the rate.	Convert units first.
Computing $y2 - y1$ only	Slope also depends on the x-step.	Divide by $x2 - x1$.
Calling any increasing table linear	Linear means constant increase, not merely increasing.	Calculate first differences.
Writing x^2 as 2x	Squaring means x times itself.	Example: 5^2 = 25, not 10.
Comparing differences with unequal x-steps	Raw differences are not comparable.	Use slope for each interval.

The 20-Second Check

- What is the input x?
- What is the output y?
- What are the units?
- Is there an initial value b?
- Is the rate constant or changing?
- Does the final answer make physical sense?

Memory line

Linear: same input step -> same output change. Non-linear: same input step -> changing output change.

10. Practice Set

Try these without looking at the answer key. Show the formula, substitution, calculation, and unit.

1. A car travels at 60 km/h for 5 hours from position 0. How far does it travel?
2. A cart begins 40 meters from a marker and moves at 12 m/s. Write its model and find its position after 7 seconds.
3. For y = 20x + 100, when does y reach 460 meters?
4. A table contains (0, 50), (2, 80), and (4, 110). Find m and b, then write the model.
5. Classify y-values 6, 10, 14, 18 when x increases by 1. Show the differences.
6. Classify delays 5, 7, 11, 17, 25 when the input increases equally. Show the differences.
7. For D(x) = x^2, find D(8). Then find the increase from D(7) to D(8).
8. Convert 72 km/h to m/s.
9. At 50 frames/second, positionSmooth = 8. What Lerp blend fraction is used for one frame? If the gap is 5 units, approximately how far is the first correction?
10. A car starts at 300 meters and moves toward the checkpoint at 15 m/s. Write the model and find its position after 8 seconds.

Scoring guide

One point for the correct model, one for substitution, one for arithmetic, and one for a correctly labeled final answer.

11. Answer Key - Problems 1 to 5

Problem 1

d = 60 x 5 = 300 km.

Problem 2

Model: y = 12x + 40. At x = 7: y = 12(7) + 40 = 84 + 40 = 124 meters.

Problem 3

460 = 20x + 100; 360 = 20x; x = 18 seconds.

Problem 4

m = (80 - 50) / (2 - 0) = 30/2 = 15. Since y = 50 when x = 0, b = 50. Model: y = 15x + 50.

Problem 5

10 - 6 = 4, 14 - 10 = 4, 18 - 14 = 4. The first difference is constant, so the relationship is linear.

12. Answer Key - Problems 6 to 10

Problem 6

Differences: +2, +4, +6, +8. They change, so the relationship is non-linear.

Problem 7

D(8) = 8^2 = 64. D(7) = 49, so the increase is 64 - 49 = 15.

Problem 8

72 / 3.6 = 20 m/s.

Problem 9

Time.deltaTime = 1/50 = 0.02 s. Blend = 8 x 0.02 = 0.16, or 16%. First correction = 5 x 0.16 = 0.8 unit.

Problem 10

Moving toward the checkpoint makes the slope negative: y = -15x + 300. At x = 8: y = -15(8) + 300 = -120 + 300 = 180 meters.

Final One-Minute Review

- Constant rate -> linear -> y = mx + b.
- Changing first differences -> non-linear.
- x^2 means x multiplied by itself.
- km/h to m/s -> divide by 3.6.
- Always label the final answer with units.

Source: CS402 - Modelling and Simulation - MIDTERM.pdf, slides 1-102. Reviewer wording and additional examples were created for study use.